



DO PROXIES ACTUALLY MATTER?

A measurable demonstration using real, reproducible rate-limit data

Testbed: GitHub REST API — unauthenticated rate limits

The Question We're Actually Testing

If I don't use a proxy, what is the real, measured impact?



Why GitHub's API

- Documented, honest limit: exactly

60 requests / hour per unauthenticated IP

Every response reports remaining quota in the

X-RateLimit-Remaining

header

No guessing, no simulation — just real numbers



Impact 1 — Identity & Location

What the target site learns about who and where you are.



Impact 2 — Rate Limiting / Blocking

How quickly a single IP gets blocked, and how proxies avoid it.

Why Rotation Works: Independent Quota Buckets

Each IP gets its own 60-request bucket. Rotating IPs means rotating buckets.

Without Proxy



requests / hour
(one IP, one bucket)

Bucket empties fast under real scraping load

VS

With Proxy Rotation (3 IPs)



Proxy 1



Proxy 2



Proxy 3

180 requests / hour effective capacity — 3x the ceiling

If one IP is throttled or banned, the other two keep working

The Blocking Test: 60+ Requests, Two Ways

Same request volume, counted as succeeded (200) vs blocked (403)



Without a proxy, the 61st request onward gets blocked the moment the hourly ceiling is hit.



With rotation across 3 IPs, no single bucket reaches its limit — zero blocks.

Note: each IP's quota resets 1 hour after its first request — run once per hour for clean results.

The Impact of Not Using a Proxy

Less anonymity, far less capacity, and much earlier IP blocking.

	Without a Proxy	With Proxy Rotation
Identity	Real IP, city & ISP exposed	Rented IP in another city
Quota	1 bucket (e.g. 60 req/hr)	N buckets (3x+ capacity)
Blocking	Blocked (403) once the limit hits	Load spread — no IP hits its limit
Resilience	One ban stops the whole scraper	One ban still leaves others running

Bottom line: process_1.ipynb (Flipkart scraping) and process_2.ipynb (rotation demo) route through proxies for exactly these measured reasons.